

WEEK 2

Week 2A — File Copy Using System Calls

This program copies the contents of one file to another using `open()`, `read()`, `write()`, and `close()`.

It demonstrates how file operations move between user space and kernel space.

```
sri_dhruthi@Dhruthi: ~  
GNU nano 8.7.1 week2a.c *  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main()  
{  
    int src, dest;  
    char buffer[1024];  
    int bytes;  
  
    src = open("input.txt", O_RDONLY);  
  
    if (src < 0)  
    {  
        printf("Cannot open source file\n");  
        return 1;  
    }  
  
    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    while ((bytes = read(src, buffer, sizeof(buffer))) > 0)  
    {  
        write(dest, buffer, bytes);  
    }  
  
    close(src);  
    close(dest);  
  
    printf("File copied successfully.\n");  
  
    return 0;  
}
```

```
sri_dhruthi@Dhruthi: ~  
sri_dhruthi@Dhruthi:~$ nano week2a.c  
sri_dhruthi@Dhruthi:~$ nano input.txt  
sri_dhruthi@Dhruthi:~$ gcc week2a.c -o week2a  
sri_dhruthi@Dhruthi:~$ ./week2a  
File copied successfully.  
sri_dhruthi@Dhruthi:~$ cat output.txt  
Operating System Lab  
File Copy using System Calls  
  
sri_dhruthi@Dhruthi:~$ |
```

Week 2B — Trace System Calls Using strace

This program/task uses strace to observe the system calls made when cat sample.txt executes.

It helps identify calls such as openat(), read(), write(), close(), and execve().

```
sri_dhruthi@Dhruthi:~$ nano sample.txt
sri_dhruthi@Dhruthi:~$ gcc week2a.c -o week2a
sri_dhruthi@Dhruthi:~$ ./week2a
File copied successfully.
sri_dhruthi@Dhruthi:~$ cat output.txt
Operating System Lab
File Copy using System Calls
```

```
sri_dhruthi@Dhruthi:~$ strace -o trace.txt cat sample.txt
Operating System Practical
File Copy using System Calls
open read write close
This file is used as input.
sri_dhruthi@Dhruthi:~$ cat trace.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffe76cd34e8 /* 26 vars */)
= 0
brk(NULL)                               = 0x56f977d20000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x77f4492e5000
```